A COMPLETE PROOFS

This supplement to the PVLDB submission *Release Determinacy for Hyperedge-Level Fairness Certification in Attributed Hypergraph Clustering* gives the complete proofs of the results whose proofs the submission shortens or sketches in Appendix A and further verdicts in Appendix B. Numbers of theorems, lemmas, propositions, corollaries, equations, figures, and tables refer to the submission, and the references are listed at the end of this supplement.

A.1 Proof of Lemma 1

Proof. Let $k = |e|$ be the number of members of e. For a group g and a cluster C, let $n_{11} = n_{e,g,C}$, and let n_{10}, n_{01}, and n_{00} count the members of e that are in g but outside C, in C but outside g, and in neither, with $k = n_{11}+n_{10}+n_{01}+n_{00}$. By Eq. (3), $\psi_e = \frac{1}{2k^2}\sum_{g,C}|R_{gC}|$ with the residual $R_{gC} = kn_{11}-(n_{11}+n_{10})(n_{11}+n_{01}) = n_{11}n_{00}-n_{10}n_{01}$. As $n_{11}+n_{10}$ depends only on g and $n_{11}+n_{01}$ only on C, we have $\sum_{g,C}(n_{11}+n_{10})(n_{11}+n_{01}) = k\cdot k = \sum_{g,C} kn_{11}$. The values R_{gC} sum to 0, and $R_{gC} = -n_{10}n_{01} \le 0$ whenever $n_{11} = 0$. It follows that $\sum_{g,C}|R_{gC}| = 2\sum_{g,C}\max(R_{gC},0) \le 2\sum_{g,C} n_{11}n_{00}$, given that $n_{10}n_{01} \ge 0$, and each term is tight when $n_{11}n_{00} = 0$ or $n_{10}n_{01} = 0$. A member in group g and cluster C differs in both group and cluster from exactly n_{00} members, so $\sum_{g,C} n_{11}n_{00}$ counts each such pair twice and equals $2N_e$. This proves $\psi_e \le 2N_e/k^2$. If the clusters inside e are a function of the groups, then $n_{10} = 0$ whenever $n_{11} > 0$, and if the groups are a function of the clusters, then $n_{01} = 0$ whenever $n_{11} > 0$, so equality holds in both cases. A cell with $n_{11}n_{00} > 0$ and $n_{10}n_{01} > 0$ needs four members, so equality also holds if $k \le 3$. □

A.2 Proof of Corollary 3

Proof. (i) The equations of the hyperedges of two members force $\eta_{uv} = \chi_{uv}/\omega(2)$ for $\mathrm{vol}(\mathcal{H})\,\mathrm{HGP}$ and $\eta_{uv} = 2/\omega(2)$ for $\mathrm{vol}(\mathcal{H})$, and the equation of f then reads $(p/q)N_f = 2N_f/3$ and $6p/q = 3$, respectively, which are the exact conditions by Theorem 1(a) and (b). Both families give each pair of f the weight $p\,\omega(2)$. The HGP of q copies of f is $\psi_f = 2N_f/9$, and that of p copies of each pair of f is $N_f/6$, as the three pairs have equal weight and each crossing pair has $\psi = 1/2$.

(ii) The hyperedge f and its three pairs have the same unweighted clique and the HGP values $2N_f/9$ and $N_f/6$ of (i). On four or more nodes, take a crossing pair ut, which exists given that f contains one, and two further nodes v and w. The inputs $\{uvw, ut, vt, wt\}$ and $\{uvt, uwt, vwt\}$ have the same normalized clique, as each pair inside uvw receives $1/2$ and each pair at t receives 1 from either side. Their values of $\mathrm{vol}(\mathcal{H})\,\mathrm{HGP}$ differ by one third of the number of crossing pairs between t and uvw, which is at least one, and the normalized clique fixes $\mathrm{vol}(\mathcal{H})$. The two inputs above are the case $uvw = 012$ and $t = 3$. □

A.3 Proof of Proposition 3(b)

Proof. (b) A hyperedge e adds $N_e/(|e|-1)$ to $\sum_{uv} w_{uv}\chi_{uv}$ and $|e|^2 I_e/(2|e|-2)$ to $\sum_{uv} w_{uv}\xi_{uv}$, whereas $\mathrm{vol}(\mathcal{H})\,\mathrm{HGP} \le \sum_e 2N_e/|e|$ by Lemma 1 and $\mathrm{vol}(\mathcal{H})\,\bar{I} = \sum_e |e|\,I_e$. As $(k-1)/k$ increases in k, $\sum_e 2N_e/|e| \le 2\frac{\bar{r}-1}{\bar{r}}\sum_{uv} w_{uv}\chi_{uv}$ and $\sum_e |e|\,I_e \ge 2\frac{r_{\min}-1}{r_{\min}}\sum_{uv} w_{uv}\xi_{uv}$, and dividing gives (b). For the constants, $U_P = \mathrm{HGP}$ on H_1, and one hyperedge of $\bar{r}$ members whose groups are its clusters, together with k hyperedges of $r_{\min}$ members that lie in one cluster and hold two groups, gives a ratio of the two sides that tends to c as k grows, and c tends to 2 as $\bar{r}$ grows with $r_{\min} = 2$. □

A.4 Proof of Theorem 3

With $s = \lceil r/2\rceil$ and $t = \lfloor r/2\rfloor$, the gap of Theorem 3 is $\gamma(r) = st(r-2)/(r^2(r-1))$. For even r, $\gamma(r) = (r-2)/(4(r-1))$, and for odd r, $\gamma(r) = (r+1)(r-2)/(4r^2)$. Any estimator from the clique returns one value on both inputs, so one error is at least $\gamma(r)/2$.

Derivation. Split the r nodes into blocks of $s = \lceil r/2\rceil$ and $t = \lfloor r/2\rfloor$ nodes, where the first forms group a and cluster C_1 and the second forms group b and cluster C_2. The first hypergraph has one hyperedge with all r nodes, and the second has the $\binom{r}{2}$ pairs as hyperedges, and both have the unweighted clique K_r.

Single hyperedge. The hyperedge has $q_e = (s/r, t/r)$, $q_{e,a} = (1,0)$, and $q_{e,b} = (0,1)$, which gives $\mathrm{TV}(q_{e,a}, q_e) = t/r$ and $\mathrm{TV}(q_{e,b}, q_e) = s/r$. With $p_{a|e} = s/r$ and $p_{b|e} = t/r$ this gives $\psi_e = \frac{s}{r}\cdot\frac{t}{r} + \frac{t}{r}\cdot\frac{s}{r} = \frac{2st}{r^2}$, and since there is one hyperedge, $\mathrm{HGP} = 2st/r^2$.

All pairs. A same-group pair lies in one cluster, so each of its groups is spread like the hyperedge and $\psi = 0$. A cross-group pair has its members in different clusters, so $q_e = (1/2, 1/2)$, each group is a point mass, both total-variation terms equal $1/2$, and $\psi = 1/2$. Each pair has weight $2/(r(r-1))$ and there are st cross-group pairs, giving $\mathrm{HGP} = st\cdot\frac{1}{2}\cdot\frac{2}{r(r-1)} = \frac{st}{r(r-1)}$.

Difference. The two values differ by $\frac{2st}{r^2} - \frac{st}{r(r-1)} = \frac{st\,[2(r-1)-r]}{r^2(r-1)} = \gamma(r)$. Evaluating at $r = 3, 4, 32$ gives the stated values, and since $st/r^2 \to 1/4$ and $(r-2)/(r-1) \to 1$, the difference tends to $1/4$.

A.5 Proof of Theorem 4

Proof. Let μ_{uv} count the hyperedges that contain u and v, and d_v the hyperedges that contain v. The release fixes μ in both parts. A compatible input has hyperedges of at most R members, where $R = r$ in (a), so it has at most $\sum_{k\le R}\binom{n}{k}$ distinct hyperedges, each with a multiplicity at most μ_{uv} for every pair uv in it. The set of compatible inputs is finite and $h_{\max}$ is attained. One compatible input, listed by its distinct hyperedges and multiplicities, is a certificate of polynomial size whose release and HGP are checked exactly, which gives membership in NP, also for $h_{\max} = U_P$, given that $\mathrm{HGP} \le U_P$ by Proposition 2(a).

Part (a). We reduce from completing a partial Latin square with $\ell \ge 1$ empty cells [21], as a complete square maps to a fixed yes-instance. Its defect graph $G = (V_G, E_G)$ has the parts V_1, V_2, V_3 of rows, columns, and symbols, joins row i to column j if cell ij is empty, and joins each row or column to the symbols that it lacks, so it has ℓ edges between any two parts. A triangle has one node in each part and fills one cell, and the square can be completed exactly when the edges of G partition into triangles. The reduction gives every edge of G that meets V_1 the same excess of U_P over HGP, and a hyperedge that covers a triangle removes the excess of two such edges, so $h_{\max}$ reaches U_P exactly when the triangles chosen partition the edges. We order V_G. For each edge st with $s < t$, we add nodes x_{st}, x_{ts} and the hyperedge $s\,t\,x_{st}\,x_{ts}$, and for each triangle uvw with $u < v < w$, private nodes $z, f_0, \ldots, f_5$ and, for each of its edges st with $s < t$, private nodes $p, q_1, q_2, o_1, o_2, y_{st}, y_{ts}$. We write $Y_{st} = s\,t\,y_{st}\,y_{ts}$ and $\mathcal{A} \equiv \mathcal{B}$ if two families of hyperedges cover the same pairs, each once. Every edge st of a triangle uvw satisfies the switch identity in the first two lines below, and the triangle

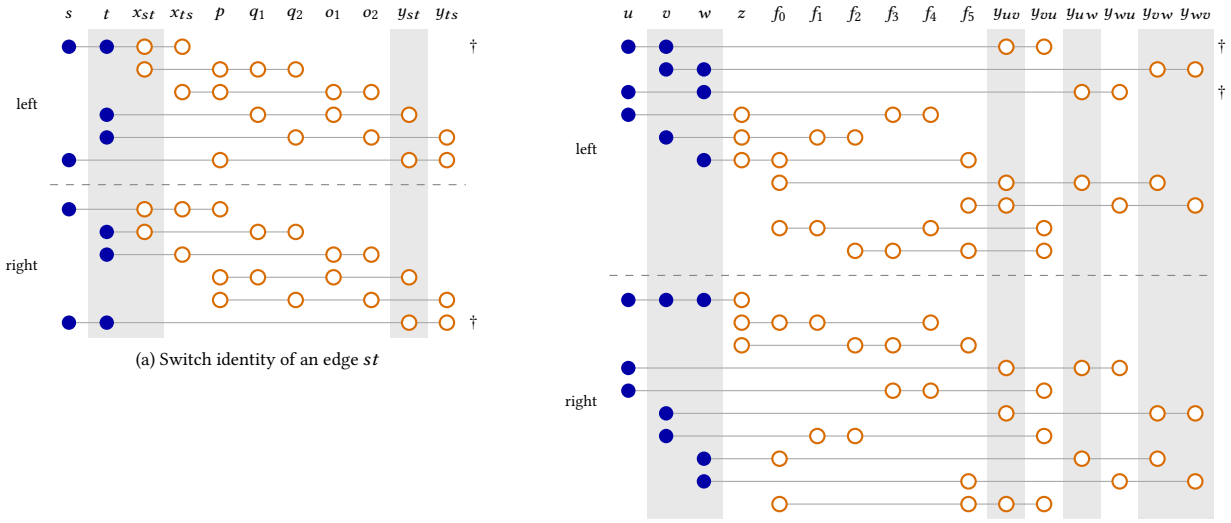


Figure 5: Gadgets of the reduction in the proof of Theorem 4(a) at $r = 4$. Columns are nodes, filled for group a and hollow for group b, with cluster C_2 shaded, and rows are hyperedges. The rows above the dashed line form the left side of an identity and the rows below it the right side, and both sides cover the same pairs, each once. (a) The switch of an edge st with $s \in V_1$ and $t \in V_2$, whose left side lies in $\mathcal{H}_0$. (b) The absorber of a triangle uvw with $u \in V_1$, whose left side consists of the hyperedges Y_{uv}, Y_{vw}, Y_{uw} that the switches produce and of seven hyperedges of $\mathcal{H}_0$. A dagger marks a hyperedge whose slack $U_P - \mathrm{HGP}$ contributes $1/4$, and every other hyperedge contributes 0.

satisfies the absorber identity in the rest. Both identities are drawn in Figure 5.

$$
\begin{aligned}
&s\,t\,x_{st}\,x_{ts},\ x_{st}\,p\,q_1\,q_2,\ x_{ts}\,p\,o_1\,o_2,\ t\,q_1\,o_1\,y_{st},\ t\,q_2\,o_2\,y_{ts},\ s\,p\,y_{st}\,y_{ts}\\
&\quad\equiv s\,x_{st}\,x_{ts}\,p,\ t\,x_{st}\,q_1\,q_2,\ t\,x_{ts}\,o_1\,o_2,\ p\,q_1\,o_1\,y_{st},\ p\,q_2\,o_2\,y_{ts},\ Y_{st},\\
&Y_{uv},\ Y_{vw},\ Y_{uw},\ u\,z\,f_3\,f_4,\ v\,z\,f_1\,f_2,\ w\,z\,f_0\,f_5,\ y_{uv}\,y_{uw}\,y_{vw}\,f_0,\\
&y_{uv}\,y_{wu}\,y_{wv}\,f_5,\ y_{vu}\,f_0\,f_1\,f_4,\ y_{vu}\,f_2\,f_3\,f_5 \equiv u\,v\,w\,z,\ z\,f_0\,f_1\,f_4,\\
&z\,f_2\,f_3\,f_5,\ u\,y_{uv}\,y_{uw}\,y_{wu},\ u\,y_{vu}\,f_3\,f_4,\ v\,y_{uv}\,y_{vw}\,y_{wv},\ v\,y_{vu}\,f_1\,f_2,\\
&w\,y_{uw}\,y_{vw}\,f_0,\ w\,y_{wu}\,y_{wv}\,f_5,\ y_{uv}\,y_{vu}\,f_0\,f_5.
\end{aligned}
$$

The given compatible input is the witness $\mathcal{H}_0$, which holds every hyperedge $s\,t\,x_{st}\,x_{ts}$, the other five left hyperedges of every switch identity, and the seven left hyperedges other than Y_{uv}, Y_{vw}, Y_{uw} of every absorber identity, and m is its number of hyperedges. The graph G has at most $n\ell$ triangles for a square of order n, so the construction takes polynomial time. Group a is V_G and group b holds the new nodes. A node x_{st} or y_{st}, in either order of s and t, takes the cluster of its second index, which places x_{st} and y_{st} in the cluster of t and x_{ts} and y_{ts} in the cluster of s. Cluster C_2 holds $V_2 \cup V_3$ and every node x_{st} or y_{st} whose second index lies in $V_2 \cup V_3$, and C_1 holds every other node. We release the weighted clique or P of $\mathcal{H}_0$ and set $B = U_P - 1/(4m)$.

Every hyperedge of $\mathcal{H}_0$ other than $s\,t\,x_{st}\,x_{ts}$ has at most one member in V_G or among the nodes x_{st}, its other members are private to one triangle, and the hyperedges built for one triangle pairwise share at most one node. This gives $\mu_{uv} \le 1$. As $w_{uv} = \omega(4)\mu_{uv}$, $P_{uv} = \mu_{uv}/4$, and $P_{vv} = \sum_u \mu_{uv}/12$, either release fixes μ, and the compatible inputs are the partitions of $\Gamma = \{uv : \mu_{uv} = 1\}$ into 4-cliques. Here Γ restricted to V_G is G. The common neighbors of an edge st outside V_G are x_{st}, x_{ts} and the nodes y_{st}, y_{ts}, z of each triangle through st, and among them the adjacent pairs are $x_{st}x_{ts}$ and the pairs $y_{st}y_{ts}$ of one triangle. Among these common neighbors, a node adjacent to a third node $w \in V_G$ is the node z of the triangle stw. A 4-clique with two nodes of V_G is $s\,t\,x_{st}\,x_{ts}$ or some Y_{st}, one with three is $uvwz$, and none has four, as G is tripartite. All 4-cliques of Γ are listed in Lemma 2 after this proof.

For a hyperedge with α_1, α_2 members of group a and β_1, β_2 of group b in C_1, C_2, every entry of Eq. (3) is $\pm(\alpha_1\beta_2 - \alpha_2\beta_1)/r^2$ and $N_e = \alpha_1\beta_2 + \alpha_2\beta_1$. With $U_P = \frac{1}{m}\sum_e 2N_e/r^2$ from the proof of Proposition 2, we obtain $U_P - \mathrm{HGP} = \frac{1}{m}\sum_e \sigma_e$ with $\sigma_e = 4\min(\alpha_1\beta_2, \alpha_2\beta_1)/r^2$, which vanishes if one group of the hyperedge lies in one cluster. A 4-clique with at most one node of V_G has at most one member of group a, and the hyperedges $uvwz$, $s\,t\,x_{st}\,x_{ts}$, and Y_{st} with $s, t \in V_2 \cup V_3$ have one group inside one cluster, so all of these have $\sigma_e = 0$. The 4-cliques with $\sigma_e > 0$ are exactly the hyperedges $s\,t\,x_{st}\,x_{ts}$ and Y_{st} whose edge st is one of the 2ℓ edges incident to V_1, and they have $\alpha_1 = \alpha_2 = \beta_1 = \beta_2 = 1$ and $\sigma_e = 1/4$. A hyperedge $uvwz$ covers two such edges, and the triangles of the λ hyperedges $uvwz$ of a compatible input are edge-disjoint, so $\mathrm{HGP} = U_P - (\ell - \lambda)/(2m)$. Conversely, for edge-disjoint triangles, replacing the left sides of their switch and then absorber identities in $\mathcal{H}_0$ by the right sides uses each hyperedge $s\,t\,x_{st}\,x_{ts}$ at most once and preserves μ. This gives $h_{\max} = U_P - (\ell - \nu)/(2m)$ for the largest number ν of edge-disjoint triangles, which is $U_P > B$ if the square can be completed and at most $U_P - 1/(2m) < B$ otherwise.

For $r \ge 5$, we add the same set Z of $r - 4$ nodes of group b in C_1 to every hyperedge of $\mathcal{H}_0$ and set $B = U_P - 4/(r^2 m)$. The release fixes m through $\sum_{u<v} \mu_{uv} = m\binom{r}{2}$, and for $z' \in Z$ the multiplicities

$\mu_{z'v}$ sum to $(r-1)m$, which is $r-1$ times the degree of z' in a compatible input, so every hyperedge contains Z. Removing Z gives a 4-clique partition of the graph Γ of the unpadded $\mathcal{H}_0$, and padding one reproduces μ, as $\mu_{z'v} = d_v = \sum_{u \notin Z} \mu_{uv}/3$. Padded crossing hyperedges have $(\alpha_1, \alpha_2, \beta_1, \beta_2) = (1, 1, r-3, 1)$ and $\sigma_e = 4/r^2$, and the other hyperedges keep $\sigma_e = 0$, so HGP $= U_P - 8(\ell - \lambda)/(r^2 m)$ and B separates the two cases. For every $r \geq 4$, a bound U with $h_{\max} \leq U < h_{\max} + 8/(r^2 m)$ satisfies $U \geq U_P$ exactly when the square can be completed.

Part (b). We reduce from partitioning the edges of a graph $G = (V_G, E_G)$ into copies of K_R [45], with $M' = |E_G| \geq 1$, as an edgeless graph maps to a fixed yes-instance. We place V_G in group a and C_1, add M' disjoint pairs $x_i y_i$ with x_i in group a and C_1 and y_i in group b and C_2, and release $\omega(2)$ on every edge of G and pair $x_i y_i$. As $\omega(k) = \omega(2)$ for $k \leq R$, every released pair lies in exactly one hyperedge and no other pair lies in a hyperedge. The compatible inputs consist of the pairs $x_i y_i$ and of cliques of two to R nodes of G that partition E_G, and the input with one size-2 hyperedge per pair is among them. Hyperedges inside V_G have $\psi_e = 0$ and each pair $x_i y_i$ has $\psi_e = 1/2$, so HGP $= M'/(\mathrm{vol}_G + 2M')$ with $\mathrm{vol}_G = \frac{2M'}{R-1} + \sum_e \frac{|e|(R-|e|)}{R-1}$ over the cliques, where a term vanishes if $|e| = R$ and is at least one otherwise, as $(|e|-1)(R-|e|-1) \geq 0$. With $S^* = 2M'R/(R-1)$, which satisfies $S^* \leq 3M'$ and $S^* + 1 \leq 4M'$, we have $h_{\max} = M'/S^*$ if G has a K_R-partition and $h_{\max} \leq M'/(S^*+1)$ otherwise. The budget $B = 2M'/(2S^*+1)$ separates the cases, and the gap $M'/(S^*(S^*+1))$ is at least $1/(12M') = 1/(6M)$ for the $M = 2M'$ released pairs, which gives the bound on U as in (a). □

We next add the details that this proof states without derivation.

4-cliques of Γ. The proof needs only the 4-cliques with two or more nodes of V_G and the fact that every other 4-clique has at most one member of group a, and the next lemma lists all 4-cliques.

Lemma 2 (4-cliques of the pair graph). *In the reduction of part (a) at $r = 4$, the 4-cliques of $\Gamma = \{uv : \mu_{uv} = 1\}$ are exactly the hyperedges that occur on either side of a switch identity or an absorber identity. A 4-clique with two nodes of V_G is a hyperedge $s\ t\ x_{st}\ x_{ts}$ or a hyperedge Y_{st} of a triangle through st, a 4-clique with three nodes of V_G is a hyperedge $uvwz$, no 4-clique has four, and every other 4-clique has at most one node of V_G.*

Proof. Call the nodes of V_G and the nodes x_{st} shared, and every other node private to the triangle for which it is added. Every hyperedge of $\mathcal{H}_0$ other than $s\ t\ x_{st}\ x_{ts}$ has at most one shared member, and its other members are private to one triangle. Hence two shared nodes are adjacent in Γ only inside a hyperedge $s\ t\ x_{st}\ x_{ts}$, two private nodes only if they are private to the same triangle, and a node private to a triangle uvw only to private nodes of uvw and to the shared nodes u, v, w and x of the three edges of uvw.

Let Q be a 4-clique of Γ. If all nodes of Q are shared, its pairs lie in hyperedges $s\ t\ x_{st}\ x_{ts}$. A node x_{st} lies in one such hyperedge, so Q is that hyperedge if it contains a node x, and otherwise Q is a 4-clique of G, which does not exist as G is tripartite. If Q contains a node private to a triangle T, all nodes of Q are private to T or shared nodes of T, so Q lies in the closed gadget of T, which consists of the three nodes of T, the six nodes x of its edges, and its 28 private nodes. On this set, Γ consists of the pairs of the three hyperedges $s\ t\ x_{st}\ x_{ts}$ of the edges of T and of the 22 hyperedges of $\mathcal{H}_0$ built for T, as a hyperedge built for another triangle meets this set in at most one shared node. Up to the names of its nodes, this graph is the pair graph of the reduction for the partial Latin square of order one with its cell empty, whose witness consists of exactly these 25 hyperedges. Exhaustive enumeration on this graph with 37 nodes and 150 pairs finds 53 4-cliques, which are the three hyperedges $s\ t\ x_{st}\ x_{ts}$, the five other left hyperedges and the six right hyperedges of each switch identity, among them the hyperedges Y_{st}, and the seven left hyperedges of $\mathcal{H}_0$ and the ten right hyperedges of the absorber identity. Conversely, each hyperedge of an identity is a 4-clique of Γ, as the two sides of an identity cover the same pairs and the left sides cover pairs of Γ. The counts of nodes of V_G in these hyperedges give the second statement. □

Smallest instances. For the partial Latin square of order one with its cell empty, G is one triangle, so $\ell = 1$, and $\mathcal{H}_0$ has $m = 25$ hyperedges on 37 nodes. Exhaustive enumeration in exact arithmetic finds 9 partitions of Γ into 4-cliques, whose HGP values are $\mathrm{HGP}(\mathcal{H}_0) = 23/200$ and $U_P = 27/200$, so $h_{\max} = U_P$ and $U_P - \mathrm{HGP}(\mathcal{H}_0) = \ell/(2m) = 1/50$. For the square of order two with one filled cell, $\ell = 3$ and $m = 97$, and the 2,261 partitions take exactly the four values $U_P - (3-\lambda)/(2m)$ for $\lambda = 0, \ldots, 3$ hyperedges $uvwz$. Lemma 2 and $U_P - \mathrm{HGP}(\mathcal{H}_0) = \ell/(2m)$ also hold on three further squares of orders two and three.

Source problem of part (a). A partial Latin square of order n is an $n \times n$ array whose filled cells hold symbols from $\{1, \ldots, n\}$, with no symbol repeated in a row or a column, and deciding whether it can be completed to a Latin square is NP-complete [21]. A triangle ijk of the defect graph has one node in each part and fills the empty cell ij with a symbol k that row i and column j both lack, so sets of edge-disjoint triangles correspond to conflict-free partial fillings. A row lacks as many symbols as it has empty cells, and so does a column.

Size of the witness. Each side of the switch identity covers 36 pairs and each side of the absorber identity covers 60. For τ triangles, $\mathcal{H}_0$ has $|E_G| + 22\tau$ hyperedges on $|V_G| + 2|E_G| + 28\tau$ nodes, where the count includes nodes of V_G that lie in no hyperedge, and $h_{\min} = \mathrm{HGP}(\mathcal{H}_0) = U_P - 8\ell/(r^2 m)$, which is $U_P - \ell/(2m)$ at $r = 4$.

Linearity of the witness. Every hyperedge of $\mathcal{H}_0$ other than $s\ t\ x_{st}\ x_{ts}$ contains at most one node of V_G or of the nodes x_{st}, and its other members are private to one triangle. Two hyperedges therefore share at most one node unless both are built for the same triangle, and two hyperedges $s\ t\ x_{st}\ x_{ts}$ share at most one endpoint. For one triangle, the six hyperedges on the left of a switch identity pairwise share exactly one node, hyperedges built for different edges share at most a node of V_G, and the seven absorber hyperedges of $\mathcal{H}_0$ pairwise share at most one node. The switch hyperedges contain no node z or f_i, and none of the seven absorber hyperedges of $\mathcal{H}_0$ contains both y_{st} and y_{ts} or a node of V_G together with a node y_{st}, so an absorber hyperedge and a switch hyperedge share at most one node. Hence $\mu_{uv} \leq 1$ on $\mathcal{H}_0$.

Neighbors in Γ. A node of V_G is adjacent in Γ to the nodes x of its edges, to the nodes p, y_{st}, y_{ts} of the switches in which it is the smaller endpoint s, to the nodes $q_1, q_2, o_1, o_2, y_{st}, y_{ts}$ of the switches

in which it is the larger endpoint t, and to z and two nodes f_i of each of its triangles, where the three nodes of a triangle receive disjoint pairs of nodes f_i. Among the common neighbors of an edge st outside V_G, no hyperedge of $\mathcal{H}_0$ contains a node x together with a node y or z, a node z together with a node y, or nodes private to different triangles, so the adjacent pairs are those stated in the proof.

Remarks on part (a). On these instances, $\mu_{uv} \le 1$ for every pair outside the added nodes forbids repeated hyperedges, so the theorem also holds for r-uniform multihypergraphs. An algorithm that decides the problem without the witness also decides it with the witness, so the hardness persists when the compatible input is withheld and only its existence is promised.

Part (b) and the random-walk release. The reduction of part (b) relies on pair counts leaving the incidence volume open, whereas the entries of P sum to $\mathrm{vol}(\mathcal{H})$ and the weights of the normalized clique sum to $\mathrm{vol}(\mathcal{H})/2$.

Example of part (b). As an instance of part (b) with $R = 3$, let G be a triangle, so $M' = 3$, $S^* = 9$, and $B = 6/19$. The compatible inputs are the triangle as one hyperedge, with HGP $= 1/3 = M'/S^*$, and its three edges as hyperedges, with HGP $= 1/4$, so $h_{\max} = 1/3 > B$. For a path with two edges, $M' = 2$, $S^* = 6$, and $B = 4/13$, and its single compatible input has HGP $= 1/4 < B$.

A.6 Constant-Gap Hardness

After Theorem 4, the body states that approximating $h_{\max}$ within an additive constant that depends only on the hyperedge size is NP-hard. We state this result with its constant and prove it here.

Theorem 5 (Constant-gap hardness). *For every fixed $r \ge 4$, unless* P = NP*, no polynomial-time auditor returns on every input of Theorem 4(a) a bound U with $h_{\max} \le U < h_{\max} + \varepsilon_r$, where $\varepsilon_r = 2/(11975\, r^2)$, and none returns on every such input an estimate U with $|U - h_{\max}| < \varepsilon_r/2$. In particular, $h_{\max}$ has no polynomial-time additive approximation scheme unless* P = NP*.*

Proof. For a graph F of maximum degree three with node set V_F, $n_0 = |V_F|$, and edge set E_F, deciding whether its independence number $\mathrm{ind}(F)$ exceeds $\xi_1 n_0$ or lies below $\xi_0 n_0$ is NP-hard for the rationals $\xi_1 = \frac{2M/k+14-2\epsilon}{4M/k+32}$ and $\xi_0 = \frac{2M/k+13+2\epsilon}{4M/k+32}$ of Chlebík and Chlebíková [18, Section 4, p. 19], where $\epsilon > 0$ is arbitrarily small and M and k are parameters of the amplifier that they use, with $M/k \le 40.4$, so that $\xi_1 - \xi_0 = (1 - 4\epsilon)/(4M/k + 32) > 1/200$ for $\epsilon \le 0.007$. Let F' replace every edge of F by a path with six inner nodes. This gives $\mathrm{ind}(F') = \mathrm{ind}(F) + 3|E_F|$, as an independent set I of F extends by the second, fourth, and sixth inner node of each edge $\hat{u}\hat{v}$ counted from $\hat{u}$ if $\hat{u} \in I$ and by the first, third, and fifth otherwise, and conversely an independent set J of F' has at most three inner nodes on each path and at most two if both ends lie in J, so deleting one end of each such path from $J \cap V_F$ leaves an independent set of F with at least $|J| - 3|E_F|$ nodes.

We embed F' in $A_0 \times A_1 \times A_2$ for disjoint sets A_i such that two points agree in exactly two coordinates if the nodes are adjacent and in at most one otherwise. A node $\hat{w}$ of F gets fresh values $\hat{w}^0, \hat{w}^1, \hat{w}^2$, and its at most three edges get distinct ports in $\{0, 1, 2\}$. For an edge with port i at $\hat{u}$ and j at $\hat{v}$, choose permutations (i, i_2, i_3) and (j_1, j_2, j) of $\{0, 1, 2\}$ with $j_1 \ne i$, and walk from the point of $\hat{u}$ by changing coordinates $i, i_2, i_3, i, j_1, j_2, j$ in turn, the first four to fresh values and the last three to $\hat{v}^{j_1}, \hat{v}^{j_2}, \hat{v}^{j}$. The first six points are the inner nodes and the last is $\hat{v}$. Consecutive directions differ and no coordinate returns to an earlier value, so consecutive points agree in exactly two coordinates and points at distance two or more on a walk in at most one. The values of a node occur exactly at that node and on the walks of its edges, and an inner point carries values of at most one node, and two values of it exactly when it is the first or last inner point. Two such points at a node $\hat{w}$ lack the values of $\hat{w}$ in different coordinates and agree in one coordinate, so points on different walks agree in two coordinates only for a node and its adjacent inner point.

Let G be the tripartite graph on $A_0 \cup A_1 \cup A_2$ whose edges are the pairs of values that occur together in a point, and T^* the triangles spanned by the points. We build the instance of part (a) with $V_1 = A_0$, $V_2 = A_1$, and $V_3 = A_2$, and with the hyperedges of the switch and absorber identities only for the triangles of T^*, which gives $m = |E_G| + 22|T^*|$ hyperedges. The argument of part (a) applies to each edge and triangle separately. It shows that $\mu_{uv} \le 1$ on the witness, that the compatible inputs partition the edges of Γ into 4-cliques, and that the 4-cliques with three nodes of V_G are the hyperedges $uvwz$ with $uvw \in T^*$. The hyperedges $s\,t\,x_{st}\,x_{ts}$ and Y_{st} whose edge st meets V_1 have $\sigma_e = 1/4$ and all other hyperedges have $\sigma_e = 0$, and each hyperedge $uvwz$ covers two such edges. A compatible input with λ hyperedges $uvwz$ has HGP $= U_P - (\theta - 2\lambda)/(4m)$, where θ is the number of edges of G at V_1. The triangles of these λ hyperedges are edge-disjoint, so $\lambda \le \nu$, and edge-disjoint triangles can be switched and absorbed independently, so $h_{\max} = U_P - (\theta - 2\nu)/(4m)$ for the largest number ν of pairwise edge-disjoint triangles in T^*.

Two triangles of T^* share an edge of G exactly when their nodes are adjacent in F', and no edge lies in three triangles, as F' has no triangle. Hence $\nu = \mathrm{ind}(F) + 3|E_F|$, $|T^*| = n_0 + 6|E_F|$, $|E_G| = 3|T^*| - 7|E_F|$, and $m = 25n_0 + 143|E_F| \le 239.5\, n_0$. With $B_i = U_P - (\theta - 6|E_F| - 2\xi_i n_0)/(4m)$, we have $h_{\max} > B_1$ if $\mathrm{ind}(F) > \xi_1 n_0$, $h_{\max} < B_0$ if $\mathrm{ind}(F) < \xi_0 n_0$, and $B_1 - B_0 = (\xi_1 - \xi_0) n_0/(2m) > 1/95800$. A bound U with $h_{\max} \le U < h_{\max} + 1/95800$ then decides the source problem by testing $U \ge B_1$, and an estimate with $|U - h_{\max}| < 1/191600$ by comparing it with $(B_0 + B_1)/2$. For $r \ge 5$, padding every hyperedge with the same $r - 4$ nodes of group b in C_1 as in part (a) multiplies every slack by $16/r^2$ and keeps m, so the gap exceeds $16/(95800\, r^2) = \varepsilon_r$. □

Remarks on Theorem 5. The proof applies to r-uniform multihypergraphs as well, because $\mu_{uv} \le 1$ outside the added nodes forbids repeated hyperedges. The NP-completeness of deciding $h_{\max} = U_P$ rests on part (a) of Theorem 4. The constant results from the 22 hyperedges per triangle of T^*, from $m \le 239.5\, n_0$, and from the gap of the source problem.

B FURTHER VERDICTS AT THE SHARED BUDGET

We list in Table 7 the verdicts at the shared budget B without the two CAHC-init cells, on the ten cells of race and disability in ACS-PUMS-HH, which have less than 30% of their memberships in hyperedges with two or more groups, and for each method. Without the CAHC-init cells, both surrogates miss all six failing cells, and 11 audits are sent to exact evaluation. All ten screened cells pass B, neither surrogate errs on them, and the median ratio of U_P to true HGP is 1.017.

Table 7: Verdicts at the shared budget B on further sets of cells. The screened cells audit race and disability in ACS-PUMS-HH with the five methods, the method rows cover the 75 cells, and CE+FL is CE+Fair-Louvain.

Set of cells	Cells	Fail	U_P certifies passing ↑	$U_{Q,50}$ certifies passing ↑
All 75 cells	75	6	62 of 69	57 of 69
Without CAHC-init	73	6	62 of 67	57 of 67
Screened race and disability	10	0	10 of 10	8 of 10
AHRC	15	1	12 of 14	11 of 14
AHCKA	15	1	13 of 14	13 of 14
GRAC	15	0	15 of 15	14 of 15
CAHC	15	2	11 of 13	9 of 13
CE+FL	15	2	11 of 13	10 of 13

REFERENCES

[1] Serge Abiteboul and Oliver M. Duschka. 1998. Complexity of Answering Queries Using Materialized Views. In *Proceedings of the Seventeenth ACM SIGACT-SIGMOD-SIGART Symposium on Principles of Database Systems*. ACM Press, New York, NY, USA, 254–263. https://doi.org/10.1145/275487.275516

[2] R. Ahlswede and H. Aydinian. 2011. On Security of Statistical Databases. *SIAM Journal on Discrete Mathematics* 25, 4 (2011), 1778–1791. https://doi.org/10.1137/070689589

[3] McKane Andrus, Elena Spitzer, Jeffrey Brown, and Alice Xiang. 2021. What We Can't Measure, We Can't Understand: Challenges to Demographic Data Procurement in the Pursuit of Fairness. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. ACM, New York, NY, USA, 249–260. https://doi.org/10.1145/3442188.3445888

[4] Marcelo Arenas, Leopoldo Bertossi, and Jan Chomicki. 1999. Consistent Query Answers in Inconsistent Databases. In *Proceedings of the Eighteenth ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems*. ACM, New York, NY, USA, 68–79. https://doi.org/10.1145/303976.303983

[5] Marcelo Arenas, Leopoldo Bertossi, Jan Chomicki, Xin He, Vijay Raghavan, and Jeremy Spinrad. 2003. Scalar Aggregation in Inconsistent Databases. *Theoretical Computer Science* 296, 3 (2003), 405–434. https://doi.org/10.1016/S0304-3975(02)00737-5

[6] Abolfazl Asudeh, Zhongjun Jin, and H. V. Jagadish. 2019. Assessing and Remedying Coverage for a Given Dataset. In *2019 IEEE 35th International Conference on Data Engineering (ICDE)*. IEEE, 554–565. https://doi.org/10.1109/ICDE.2019.00056

[7] Albert Atserias, Martin Grohe, and Dániel Marx. 2013. Size Bounds and Query Plans for Relational Joins. *SIAM J. Comput.* 42, 4 (2013), 1737–1767. https://doi.org/10.1137/110859440

[8] Rachel K. E. Bellamy, Kuntal Dey, Michael Hind, Samuel C. Hoffman, Stephanie Houde, Kalapriya Kannan, Pranay Lohia, Jacquelyn Martino, Sameep Mehta, Aleksandra Mojsilović, Seema Nagar, Karthikeyan Natesan Ramamurthy, John Richards, Diptikalyan Saha, Prasanna Sattigeri, Moninder Singh, Kush R. Varshney, and Yunfeng Zhang. 2019. AI Fairness 360: An Extensible Toolkit for Detecting and Mitigating Algorithmic Bias. *IBM Journal of Research and Development* 63, 4/5 (2019), 4:1–4:15. https://doi.org/10.1147/JRD.2019.2942287

[9] Manish Anand Bhide, Pranay Kumar Lohia, Diptikalyan Saha, and Madhavi Katari. 2022. Bias Source Identification and De-Biasing of a Dataset. US Patent 11,475,331 B2, filed June 25, 2020, granted October 18, 2022, assigned to International Business Machines Corporation. https://patents.google.com/patent/US11475331B2

[10] P. J. Bickel, E. A. Hammel, and J. W. O'Connell. 1975. Sex Bias in Graduate Admissions: Data from Berkeley. *Science* 187, 4175 (1975), 398–404. https://doi.org/10.1126/science.187.4175.398

[11] Asia J. Biega, Peter Potash, Hal Daumé III, Fernando Diaz, and Michèle Finck. 2020. Operationalizing the Legal Principle of Data Minimization for Personalization. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*. ACM, New York, NY, USA, 399–408. https://doi.org/10.1145/3397271.3401034

[12] Paolo Boldi and Sebastiano Vigna. 2004. The WebGraph Framework I: Compression Techniques. In *Proceedings of the 13th International Conference on World Wide Web (WWW)*. ACM, 595–602. https://doi.org/10.1145/988672.988752

[13] Guy Bresler, Chenghao Guo, Yury Polyanskiy, and Andrew Yao. 2025. Partial and Exact Recovery of a Random Hypergraph from its Graph Projection. In *Proceedings of the Thirty Eighth Conference on Learning Theory (Proceedings of Machine Learning Research)*, Vol. 291. PMLR, 534–593.

[14] William J. Carrington and Kenneth R. Troske. 1997. On Measuring Segregation in Samples With Small Units. *Journal of Business & Economic Statistics* 15, 4 (1997), 402–409. https://doi.org/10.1080/07350015.1997.10524718

[15] Xingyu Chen, Brandon Fain, Liang Lyu, and Kamesh Munagala. 2019. Proportionally Fair Clustering. In *Proceedings of the 36th International Conference on Machine Learning (ICML) (Proceedings of Machine Learning Research)*, Vol. 97. PMLR, Long Beach, California, USA, 1032–1041.

[16] Flavio Chierichetti, Ravi Kumar, Silvio Lattanzi, and Sergei Vassilvitskii. 2017. Fair Clustering Through Fairlets. In *Advances in Neural Information Processing Systems*, Vol. 30. Curran Associates, Inc., Red Hook, NY, USA, 5029–5037.

[17] Francis Y. L. Chin and Gültekin Özsoyoğlu. 1982. Auditing and Inference Control in Statistical Databases. *IEEE Transactions on Software Engineering* SE-8, 6 (1982), 574–582. https://doi.org/10.1109/TSE.1982.236161

[18] Miroslav Chlebík and Janka Chlebíková. 2003. *Inapproximability Results for Bounded Variants of Optimization Problems.* Technical Report TR03-026. Electronic Colloquium on Computational Complexity (ECCC).

[19] Ben Chugg, Santiago Cortes-Gomez, Bryan Wilder, and Aaditya Ramdas. 2023. Auditing Fairness by Betting. In *Advances in Neural Information Processing Systems 36*. Curran Associates, Inc., Red Hook, NY, USA, 6070–6091. https://doi.org/10.52202/075280-0266

[20] Sara Cohen, Werner Nutt, and Yehoshua Sagiv. 2006. Rewriting Queries with Arbitrary Aggregation Functions Using Views. *ACM Transactions on Database Systems* 31, 2 (2006), 672–715. https://doi.org/10.1145/1138394.1138400

[21] Charles J. Colbourn. 1984. The Complexity of Completing Partial Latin Squares. *Discrete Applied Mathematics* 8, 1 (1984), 25–30. https://doi.org/10.1016/0166-218X(84)90075-1

[22] Alin Deutsch, Alan Nash, and Jeff Remmel. 2008. The Chase Revisited. In *Proceedings of the 27th ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems (PODS)*. ACM, New York, NY, USA, 149–158. https://doi.org/10.1145/1376916.1376938

[23] Adrian Dobra and Stephen E. Fienberg. 2000. Bounds for Cell Entries in Contingency Tables Given Marginal Totals and Decomposable Graphs. *Proceedings of the National Academy of Sciences* 97, 22 (2000), 11885–11892. https://doi.org/10.1073/pnas.97.22.11885

[24] Yushun Dong, Zhenyu Lei, Zaiyi Zheng, Song Wang, Jing Ma, Alex Jing Huang, Chen Chen, and Jundong Li. 2024. PyGDebias: A Python Library for Debiasing in Graph Learning. In *Companion Proceedings of the ACM Web Conference 2024*. ACM, 1019–1022. https://doi.org/10.1145/3589335.3651239

[25] Otis Dudley Duncan and Beverly Duncan. 1955. A Methodological Analysis of Segregation Indexes. *American Sociological Review* 20, 2 (1955), 210–217. https://doi.org/10.2307/2088328

[26] European Parliament and Council. 2024. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union, L Series, 2024/1689, 12 July 2024. https://eur-lex.europa.eu/eli/reg/2024/1689/oj Article 10 (Data and data governance); Annex III (high-risk AI systems).

[27] Barakeel Fanseu Kamhoua, Lin Zhang, Kaili Ma, James Cheng, Bo Li, and Bo Han. 2021. HyperGraph Convolution Based Attributed HyperGraph Clustering. In *Proceedings of the 30th ACM International Conference on Information & Knowledge Management (CIKM '21)*. Association for Computing Machinery, New York, NY, USA, 453–463. https://doi.org/10.1145/3459637.3482437

[28] Michael Feldman, Sorelle A. Friedler, John Moeller, Carlos Scheidegger, and Suresh Venkatasubramanian. 2015. Certifying and Removing Disparate Impact. In *Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. ACM, New York, NY, USA, 259–268. https://doi.org/10.1145/2783258.2783311

[29] Zijin Feng, Miao Qiao, and Hong Cheng. 2023. Modularity-Based Hypergraph Clustering: Random Hypergraph Model, Hyperedge-Cluster Relation, and Computation. *Proceedings of the ACM on Management of Data* 1, 3, Article 215 (2023), 25 pages. https://doi.org/10.1145/3617335

[30] Zijin Feng, Miao Qiao, Chengzhi Piao, and Hong Cheng. 2025. On Graph Representation for Attributed Hypergraph Clustering. *Proceedings of the ACM on Management of Data* 3, 1, Article 59 (2025), 26 pages. https://doi.org/10.1145/3709741

[31] Matthias Fey and Jan Eric Lenssen. 2019. Fast Graph Representation Learning with PyTorch Geometric. https://doi.org/10.48550/arXiv.1903.02428 arXiv:1903.02428 [cs.LG] ICLR 2019 Workshop on Representation Learning on Graphs and Manifolds.

[32] Ariel Fuxman, Elham Fazli, and Renée J. Miller. 2005. ConQuer: Efficient Management of Inconsistent Databases. In *Proceedings of the 2005 ACM SIGMOD International Conference on Management of Data*. ACM, New York, NY, USA, 155–166. https://doi.org/10.1145/1066157.1066176

[33] Mohsen Ghassemi, Alan Mishler, Niccolo Dalmasso, Luhao Zhang, Vamsi K. Potluru, Tucker Balch, and Manuela Veloso. 2025. Auditing and Enforcing Conditional Fairness via Optimal Transport. *Proceedings of the AAAI Conference on Artificial Intelligence* 39, 16 (2025), 16808–16816. https://doi.org/10.1609/aaai.v39i16.33847

[34] Christos Gkartzios, Evaggelia Pitoura, and Panayiotis Tsaparas. 2025. Modularity-Fair Deep Community Detection. In *2025 IEEE International Conference on Data Mining (ICDM)*. IEEE Computer Society, Los Alamitos, CA, USA, 287–296. https://doi.org/10.1109/ICDM65498.2025.00036

[35] Tomasz Gogacz and Jerzy Marcinkowski. 2016. Red Spider Meets a Rainworm: Conjunctive Query Finite Determinacy Is Undecidable. In *Proceedings of the 35th ACM SIGMOD-SIGACT-SIGAI Symposium on Principles of Database Systems (PODS '16)*. Association for Computing Machinery, New York, NY, USA, 121–134. https://doi.org/10.1145/2902251.2902288

[36] Daniel H. Gottlieb. 1966. A Certain Class of Incidence Matrices. *Proc. Amer. Math. Soc.* 17, 6 (1966), 1233–1237. https://doi.org/10.1090/S0002-9939-1966-0204305-9

[37] Ronald L. Graham, Shuo-Yen Robert Li, and Wen-Ching Winnie Li. 1980. On the Structure of t-Designs. *SIAM J. Algebraic Discrete Methods* 1, 1 (1980), 8–14. https://doi.org/10.1137/0601002

[38] Jack E. Graver and Wolfgang B. Jurkat. 1973. The Module Structure of Integral Designs. *Journal of Combinatorial Theory, Series A* 15, 1 (1973), 75–90. https://doi.org/10.1016/0097-3165(73)90037-X

[39] Lawrence Haddad and Ravi Kanbur. 1990. How Serious is the Neglect of Intra-Household Inequality? *The Economic Journal* 100, 402 (1990), 866–881. https://doi.org/10.2307/2233663

[40] Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. 2008. Exploring Network Structure, Dynamics, and Function using NetworkX. In *Proceedings of the 7th Python in Science Conference (SciPy 2008)*. 11–15. https://doi.org/10.25080/TCWV9851

[41] Alon Y. Halevy. 2001. Answering Queries Using Views: A Survey. *The VLDB Journal* 10, 4 (2001), 270–294. https://doi.org/10.1007/s007780100054

[42] F. Maxwell Harper and Joseph A. Konstan. 2015. The MovieLens Datasets: History and Context. *ACM Transactions on Interactive Intelligent Systems* 5, 4 (2015), 19:1–19:19. https://doi.org/10.1145/2827872

[43] Nicholas J. Higham. 2002. *Accuracy and Stability of Numerical Algorithms* (second ed.). Society for Industrial and Applied Mathematics, Philadelphia, PA. https://doi.org/10.1137/1.9780898718027

[44] Kenneth Holstein, Jennifer Wortman Vaughan, Hal Daumé III, Miro Dudík, and Hanna Wallach. 2019. Improving Fairness in Machine Learning Systems: What Do Industry Practitioners Need?. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, Article 600, 16 pages. https://doi.org/10.1145/3290605.3300830

[45] Ian Holyer. 1981. The NP-Completeness of Some Edge-Partition Problems. *SIAM J. Comput.* 10, 4 (1981), 713–717. https://doi.org/10.1137/0210054

[46] Weihua Hu, Matthias Fey, Marinka Zitnik, Yuxiao Dong, Hongyu Ren, Bowen Liu, Michele Catasta, and Jure Leskovec. 2020. Open Graph Benchmark: Datasets for Machine Learning on Graphs. In *Advances in Neural Information Processing Systems*, Vol. 33. Curran Associates, Inc., 22118–22133. https://proceedings.neurips.cc/paper_files/paper/2020/file/fb60d411a5c5b72b2e7d3527cfc84fd0-Paper.pdf

[47] IMDb. 2026. IMDb Non-Commercial Datasets. https://data.imdb.com/non-commercial-datasets/. Snapshot of 2026-05-07.

[48] Tomasz Imieliński and Witold Lipski. 1984. Incomplete Information in Relational Databases. *J. ACM* 31, 4 (1984), 761–791. https://doi.org/10.1145/1634.1886

[49] Nathan Kallus, Xiaojie Mao, and Angela Zhou. 2022. Assessing Algorithmic Fairness with Unobserved Protected Class Using Data Combination. *Management Science* 68, 3 (2022), 1959–1981. https://doi.org/10.1287/mnsc.2020.3850

[50] Raj Kamal and Amitabha Bagchi. 2024. A Lovász-Simonovits Theorem for Hypergraphs with Application to Local Clustering. *Proceedings of the ACM on Management of Data* 2, 4, Article 190 (2024), 27 pages. https://doi.org/10.1145/3677126

[51] Bogumił Kamiński, Paweł Misiorek, Paweł Prałat, and François Théberge. 2024. Modularity Based Community Detection in Hypergraphs. *Journal of Complex Networks* 12, 5 (2024), cnae041. https://doi.org/10.1093/comnet/cnae041

[52] William M. Kantor. 1972. On Incidence Matrices of Finite Projective and Affine Spaces. *Mathematische Zeitschrift* 124, 4 (1972), 315–318. https://doi.org/10.1007/BF01113923

[53] T. Karmel and M. MacLachlan. 1988. Occupational Sex Segregation—Increasing or Decreasing? *Economic Record* 64, 3 (1988), 187–195. https://doi.org/10.1111/j.1475-4932.1988.tb02057.x

[54] Michael Kearns, Seth Neel, Aaron Roth, and Zhiwei Steven Wu. 2018. Preventing Fairness Gerrymandering: Auditing and Learning for Subgroup Fairness. In *Proceedings of the 35th International Conference on Machine Learning (Proceedings of Machine Learning Research)*, Vol. 80. PMLR, Stockholm, Sweden, 2564–2572.

[55] Jon Kleinberg, Christos Papadimitriou, and Prabhakar Raghavan. 2003. Auditing Boolean Attributes. *J. Comput. System Sci.* 66, 1 (2003), 244–253. https://doi.org/10.1016/S0022-0000(02)00036-3

[56] Matthäus Kleindessner, Samira Samadi, Pranjal Awasthi, and Jamie Morgenstern. 2019. Guarantees for Spectral Clustering with Fairness Constraints. In *Proceedings of the 36th International Conference on Machine Learning (ICML) (Proceedings of Machine Learning Research)*, Vol. 97. PMLR, Long Beach, California, USA, 3458–3467. https://proceedings.mlr.press/v97/kleindessner19b.html

[57] Tarun Kumar, Sankaran Vaidyanathan, Harini Ananthapadmanabhan, Srinivasan Parthasarathy, and Balaraman Ravindran. 2020. Hypergraph clustering by iteratively reweighted modularity maximization. *Applied Network Science* 5, 1 (2020), 52. https://doi.org/10.1007/s41109-020-00300-3

[58] Yash Kurkure, Miles Shamo, Joseph Wiseman, Sainyam Galhotra, and Stavros Sintos. 2024. Faster Algorithms for Fair Max-Min Diversification in $\mathbb{R}^d$. *Proceedings of the ACM on Management of Data* 2, 3, Article 137 (2024), 26 pages. https://doi.org/10.1145/3654940

[59] Jarosław Kwiecień, Jerzy Marcinkowski, and Piotr Ostropolski-Nalewaja. 2022. Determinacy of Real Conjunctive Queries. The Boolean Case. In *Proceedings of the 41st ACM SIGMOD-SIGACT-SIGAI Symposium on Principles of Database Systems (PODS)*. ACM, New York, NY, USA, 347–358. https://doi.org/10.1145/3517804.3524168

[60] Jure Leskovec, Jon Kleinberg, and Christos Faloutsos. 2007. Graph Evolution: Densification and Shrinking Diameters. *ACM Transactions on Knowledge Discovery from Data* 1, 1, Article 2 (2007), 41 pages. https://doi.org/10.1145/1217299.1217301

[61] Yiran Li, Renchi Yang, and Jieming Shi. 2023. Efficient and Effective Attributed Hypergraph Clustering via K-Nearest Neighbor Augmentation. *Proceedings of the ACM on Management of Data* 1, 2, Article 116 (June 2023), 23 pages. https://doi.org/10.1145/3589261

[62] Leonid Libkin. 2016. Certain Answers as Objects and Knowledge. *Artificial Intelligence* 232 (2016), 1–19. https://doi.org/10.1016/j.artint.2015.11.004

[63] Jeremy Lise and Shannon Seitz. 2011. Consumption Inequality and Intra-household Allocations. *The Review of Economic Studies* 78, 1 (2011), 328–355. https://doi.org/10.1093/restud/rdq003

[64] Konstantinos Manolis and Evaggelia Pitoura. 2023. Modularity-Based Fairness in Community Detection. In *2023 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*. ACM, New York, NY, USA, 126–130. https://doi.org/10.1145/3625007.3627518

[65] Douglas S. Massey and Nancy A. Denton. 1993. *American Apartheid: Segregation and the Making of the Underclass*. Harvard University Press, Cambridge, MA.

[66] R. M. McConnell, K. Mehlhorn, S. Näher, and P. Schweitzer. 2011. Certifying Algorithms. *Computer Science Review* 5, 2 (2011), 119–161. https://doi.org/10.1016/j.cosrev.2010.09.009

[67] Evi Micha and Nisarg Shah. 2020. Proportionally Fair Clustering Revisited. In *Proceedings of the 47th International Colloquium on Automata, Languages, and Programming (ICALP) (Leibniz International Proceedings in Informatics (LIPIcs))*, Vol. 168. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, Dagstuhl, Germany, 85:1–85:16. https://doi.org/10.4230/LIPIcs.ICALP.2020.85

[68] Darie Moldovan. 2023. Algorithmic Decision Making Methods for Fair Credit Scoring. *IEEE Access* 11 (2023), 59729–59743. https://doi.org/10.1109/ACCESS.2023.3286018

[69] Alan Nash, Luc Segoufin, and Victor Vianu. 2010. Views and Queries: Determinacy and Rewriting. *ACM Transactions on Database Systems* 35, 3 (2010), 21:1–21:41. https://doi.org/10.1145/1806907.1806913

[70] M. E. J. Newman. 2001. Scientific Collaboration Networks. II. Shortest Paths, Weighted Networks, and Centrality. *Physical Review E* 64, 1, Article 016132 (2001), 7 pages. https://doi.org/10.1103/PhysRevE.64.016132

[71] M. E. J. Newman. 2001. The Structure of Scientific Collaboration Networks. *Proceedings of the National Academy of Sciences* 98, 2 (2001), 404–409. https://doi.org/10.1073/pnas.98.2.404

[72] Li Ni, Shuaikang Zeng, Lin Mu, and Longlong Lin. 2026. From Representation to Clusters: A Contrastive Learning Approach for Attributed Hypergraph Clustering. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*. Association for Computing Machinery, New York, NY, USA, 1080–1091. https://doi.org/10.1145/3774904.3792432

[73] Bryan Parno, Jon Howell, Craig Gentry, and Mariana Raykova. 2013. Pinocchio: Nearly Practical Verifiable Computation. In *2013 IEEE Symposium on Security and Privacy*. IEEE, Piscataway, NJ, USA, 238–252. https://doi.org/10.1109/SP.2013.47

[74] Evaggelia Pitoura, Kostas Stefanidis, and Georgia Koutrika. 2022. Fairness in Rankings and Recommendations: An Overview. *The VLDB Journal* 31, 3 (2022), 431–458. https://doi.org/10.1007/s00778-021-00697-y

[75] Jason Priem, Heather Piwowar, and Richard Orr. 2022. OpenAlex: A Fully-Open Index of Scholarly Works, Authors, Venues, Institutions, and Concepts. https://doi.org/10.48550/arXiv.2205.01833 arXiv:2205.01833

[76] Inioluwa Deborah Raji and Joy Buolamwini. 2019. Actionable Auditing: Investigating the Impact of Publicly Naming Biased Performance Results of Commercial AI Products. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society (AIES)*. ACM, New York, NY, USA, 429–435. https://doi.org/10.1145/3306618.3314244

[77] Inioluwa Deborah Raji, Andrew Smart, Rebecca N. White, Margaret Mitchell, Timnit Gebru, Ben Hutchinson, Jamila Smith-Loud, Daniel Theron, and Parker Barnes. 2020. Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT*)*. ACM, New York, NY, USA, 33–44. https://doi.org/10.1145/3351095.3372873

[78] Sean F. Reardon and Glenn Firebaugh. 2002. Measures of Multigroup Segregation. *Sociological Methodology* 32, 1 (2002), 33–67. https://doi.org/10.1111/1467-9531.00110

[79] James M. Sakoda. 1981. A Generalized Index of Dissimilarity. *Demography* 18, 2 (1981), 245–250. https://doi.org/10.2307/2061096

[80] Babak Salimi, Luke Rodriguez, Bill Howe, and Dan Suciu. 2019. Interventional Fairness: Causal Database Repair for Algorithmic Fairness. In *Proceedings of the 2019 International Conference on Management of Data (SIGMOD)*. ACM, New York, NY, USA, 793–810. https://doi.org/10.1145/3299869.3319901

[81] Heather Sarsons, Klarita Gërxhani, Ernesto Reuben, and Arthur Schram. 2021. Gender Differences in Recognition for Group Work. *Journal of Political Economy* 129, 1 (2021), 101–147. https://doi.org/10.1086/711401

[82] Nima Shahbazi, Stavros Sintos, and Abolfazl Asudeh. 2024. FairHash: A Fair and Memory/Time-Efficient Hashmap. *Proceedings of the ACM on Management of Data* 2, 3, Article 136 (2024), 29 pages. https://doi.org/10.1145/3654939

[83] Divesh Srivastava, Shaul Dar, H. V. Jagadish, and Alon Y. Levy. 1996. Answering Queries with Aggregation Using Views. In *Proceedings of the 22nd International Conference on Very Large Data Bases (VLDB)*. Morgan Kaufmann, San Francisco, CA, USA, 318–329.

[84] Julia Stoyanovich, Bill Howe, and H. V. Jagadish. 2020. Responsible Data Management. *Proceedings of the VLDB Endowment* 13, 12 (2020), 3474–3488.

https://doi.org/10.14778/3415478.3415570
[85] Seth Sullivant. 2005. Small Contingency Tables with Large Gaps. *SIAM Journal on Discrete Mathematics* 18, 4 (2005), 787–793. https://doi.org/10.1137/S0895480104444090
[86] Elham Tabassi. 2023. *Artificial Intelligence Risk Management Framework (AI RMF 1.0).* Technical Report NIST AI 100-1. National Institute of Standards and Technology, Gaithersburg, MD, USA. https://doi.org/10.6028/NIST.AI.100-1
[87] Luboš Takáč and Michal Zábovský. 2012. Data Analysis in Public Social Networks. In *International Scientific Conference and International Workshop Present Day Trends of Innovations*. Łomża, Poland, 110–115.
[88] Roberto Tamassia. 2003. Authenticated Data Structures. In *Algorithms – ESA 2003 (Lecture Notes in Computer Science)*, Vol. 2832. Springer, Berlin, Heidelberg, 2–5. https://doi.org/10.1007/978-3-540-39658-1_2
[89] U.S. Census Bureau. 2024. American Community Survey 2023 1-Year Public Use Microdata Sample (PUMS). https://www.census.gov/programs-surveys/acs/microdata.html.
[90] Michael Veale and Reuben Binns. 2017. Fairer Machine Learning in the Real World: Mitigating Discrimination without Collecting Sensitive Data. *Big Data & Society* 4, 2 (2017), 1–17. https://doi.org/10.1177/2053951717743530
[91] Sandra Wachter, Brent Mittelstadt, and Chris Russell. 2021. Why Fairness Cannot Be Automated: Bridging the Gap Between EU Non-Discrimination Law and AI. *Computer Law & Security Review* 41, Article 105567 (2021). https://doi.org/10.1016/j.clsr.2021.105567
[92] Xuemin Wang, Yuanjing Hao, and Liang Chang. 2026. *Supplementary Material for Release Determinacy for Hyperedge-Level Fairness Certification in Attributed Hypergraph Clustering.* Supplementary Material. Guilin University of Electronic Technology. https://github.com/yingkongshibetter/release-determinacy-ahc-pvldb20/blob/main/supplementary_material.pdf
[93] Yanbang Wang and Jon Kleinberg. 2024. From Graphs to Hypergraphs: Hypergraph Projection and its Reconstruction. In *International Conference on Learning Representations (ICLR)*. 52745–52775. https://proceedings.iclr.cc/paper_files/paper/2024/hash/e92b8441b702d142435f243806f9ec5a-Abstract-Conference.html
[94] Joyce Jiyoung Whang, Rundong Du, Sangwon Jung, Geon Lee, Barry Drake, Qingqing Liu, Seonggoo Kang, and Haesun Park. 2020. MEGA: Multi-View Semi-Supervised Clustering of Hypergraphs. *Proceedings of the VLDB Endowment* 13, 5 (2020), 698–711. https://doi.org/10.14778/3377369.3377378
[95] Siyuan Xia, Zhiru Zhu, Chris Zhu, Jinjin Zhao, Kyle Chard, Aaron J. Elmore, Ian Foster, Michael Franklin, Sanjay Krishnan, and Raul Castro Fernandez. 2022. Data Station: Delegated, Trustworthy, and Auditable Computation to Enable Data-Sharing Consortia with a Data Escrow. *Proceedings of the VLDB Endowment* 15, 11 (2022), 3172–3185. https://doi.org/10.14778/3551793.3551861
[96] Wen Xu, Pengpeng Qiao, Shang Liu, Zhirun Zheng, Yang Cao, and Zhetao Li. 2025. Continuous Publication of Weighted Graphs with Local Differential Privacy. *Proceedings of the VLDB Endowment* 18, 11 (2025), 4214–4226. https://doi.org/10.14778/3749646.3749688
[97] Jaewon Yang and Jure Leskovec. 2012. Defining and Evaluating Network Communities Based on Ground-Truth. In *2012 IEEE 12th International Conference on Data Mining (ICDM)*. IEEE, 745–754. https://doi.org/10.1109/ICDM.2012.138
[98] Jean-Gabriel Young, Giovanni Petri, and Tiago P. Peixoto. 2021. Hypergraph Reconstruction from Network Data. *Communications Physics* 4, 1 (2021), 135. https://doi.org/10.1038/s42005-021-00637-w
[99] Quan Yuan, Zhikun Zhang, Linkang Du, Min Chen, Peng Cheng, and Mingyang Sun. 2023. PrivGraph: Differentially Private Graph Data Publication by Exploiting Community Information. In *32nd USENIX Security Symposium (USENIX Security 23)*. USENIX Association, Anaheim, CA, 3241–3258. https://www.usenix.org/conference/usenixsecurity23/presentation/yuan-quan
[100] Juntao Zhang, Sheng Wang, Yuan Sun, and Zhiyong Peng. 2023. Prerequisite-Driven Fair Clustering on Heterogeneous Information Networks. *Proceedings of the ACM on Management of Data* 1, 2, Article 122 (2023), 27 pages. https://doi.org/10.1145/3589267
[101] Yupeng Zhang, Daniel Genkin, Jonathan Katz, Dimitrios Papadopoulos, and Charalampos Papamanthou. 2017. vSQL: Verifying Arbitrary SQL Queries over Dynamic Outsourced Databases. In *Proceedings of the 38th IEEE Symposium on Security and Privacy (S&P)*. IEEE, Piscataway, NJ, USA, 863–880. https://doi.org/10.1109/SP.2017.43
[102] Yupeng Zhang, Jonathan Katz, and Charalampos Papamanthou. 2015. IntegriDB: Verifiable SQL for Outsourced Databases. In *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, New York, NY, USA, 1480–1491. https://doi.org/10.1145/2810103.2813711
[103] Shengkun Zhu, Quanqing Xu, Jinshan Zeng, Sheng Wang, Yuan Sun, Zhifeng Yang, Chuanhui Yang, and Zhiyong Peng. 2023. F3KM: Federated, Fair, and Fast k-Means. *Proceedings of the ACM on Management of Data* 1, 4, Article 241 (2023), 25 pages. https://doi.org/10.1145/3626728